

Meta Analysis

An exploratory meta-analysis of the simulated impacts of temperature on sugarcane yield

Duy Khanh Nguyen - r0968831

Hoang Giang Pham - r1021358

Introduction

Sugarcane (*Saccharum officinarum* L.) is one of the world's most important cash crops, supplying the majority of global sugar production and an increasing share of bioethanol feedstock (FAO, 2023). Its cultivation supports the livelihoods of millions of smallholder farmers and rural communities across tropical and subtropical regions. However, climate change - through rising temperatures, shifting rainfall patterns, and an increasing frequency of extreme weather events - poses an escalating threat to the productivity and stability of sugarcane production systems worldwide (Zhao & Li, 2015). Developing resilient agricultural systems requires robust, up-to-date evidence about how climate change affects sugarcane yields to inform effective adaptation strategies and policy decisions.

However, assessing the impact of climate change on crop yields is challenging due to uncertainties inherent to analytical frameworks and choices. These uncertainties arise from the structure and parameterisation of crop simulation models (Asseng et al., 2013), the choice of greenhouse gas emission scenarios (White et al., 2011), and the selection of climate projections. For sugarcane, simulated yields have been shown to vary substantially depending on differences in crop model architecture (Marin et al., 2015) and the climate scenario used, e.g., A2 vs. B2 or RCP 4.5 vs. 8.5 (Marin et al., 2013). Regional heterogeneity in agro-climatic conditions, varieties, and the presence or absence of adaptation measures such as adjusted planting dates further contribute to inconsistent and sometimes contradictory findings (Challinor et al., 2014). Because of these uncertainties, projections differ not only on how much sugarcane yields will change, but sometimes on whether the effect will be positive or negative.

Meta-analysis offers a systematic means of synthesising results across independent studies, thereby quantifying central tendencies and exploring sources of heterogeneity that individual studies cannot address (Borenstein et al., 2021). By combining simulation outputs from many studies, meta-analysis can uncover the underlying climate signal despite differences in methods, locations, and scenarios. The meta-analysis by Li et al. (2021) on cotton provided a useful example of this approach, demonstrating that temperature, precipitation, CO₂ concentration, and adaptation measures all have significant, quantifiable effects on simulated yield, and that much of the variability between studies is attributable to differences in climate projections, study sites, and emission scenarios.

Drawing on data from a recent systematic review on climate change impacts on sugarcane yield conducted by researchers at the Alliance of Bioversity International and CIAT (see Literature screening), and on the analytical framework of Li et al. (2021), we apply a similar meta-analytical approach to sugarcane. We focus specifically on the effect of temperature change, which was most consistently reported across the included studies, and address two research questions:

1. What are the projected magnitudes and direction of changes in sugarcane yield under future climate conditions?
2. What is the estimated effect of each 1°C of warming on sugarcane yield?

Given the large methodological variation observed across the broader body of sugarcane climate-impact research, we deliberately restrict the meta-analysis to studies that employ process-based crop simulation models, ensuring that the included observations are as comparable as possible. The findings offer an initial quantitative summary of temperature effects on sugarcane yield from crop simulation studies, and identify key gaps that future work should address.

Methodology

Literature screening

The data used in this analysis were drawn from a broader systematic review conducted by our research group at the Alliance of Bioversity International and CIAT. The review followed PRISMA guidelines (Page et al., 2021) to ensure a rigorous, transparent, and reproducible search and screening process. We defined the research questions and devised inclusion and exclusion criteria using the PECO (Population–Exposure–Comparator–Outcome) framework (see Appendix A.2). A comprehensive search of Web of Science and ScienceDirect was performed; the full search string, target locations, filters, and the PRISMA flow diagram are provided in the Appendix A.3.

The systematic review identified 84 studies that met the inclusion criteria. However, substantial methodological heterogeneity was evident across these studies: some used statistical regression approaches, some relied on empirical field trials, and other applied process-based crop models under diverse configurations of crop models, climate projections, and emission scenarios. For this exploratory analysis, we restricted the dataset to nine studies that employed process-based crop simulation models to project future sugarcane yield under different temperature regimes (see Appendix A.1). While this ensured that the included observations were methodologically comparable, the small sample size is a limitation and a more comprehensive meta-analysis would require extending the search to include studies using other approaches. The key characteristics of the studies are listed in Table 1.

Table 1. Key characteristics of the nine studies included in the meta-analysis

Reference	Study sites	Crop model	Baseline
Verma et al. (2023)	Chhapar, India	DSSAT-CANEGRO	1979–2013
Guhan et al. (2024)	Tamil Nadu, India	DSSAT-CANEGRO	1981–2022
Sanders et al. (2019)	Honduras	AgMIP GGCMi	1960–1990
Jones et al. (2015)	La Mercy, Pongola, Malelane (South Africa)	DSSAT-CANEGRO	1980–2010
Jones & Singels (2018)	Komatipoort (South Africa), Chiredzi (Zimbabwe), Ligne Paradis (Réunion), Ayr (Australia), Piracicaba	DSSAT-CANEGRO	1980–2010

	(Brazil), La Mercy (South Africa)		
Marin et al. (2013)	Piracicaba, Ilha Solteira, São Paulo (Brazil)	DSSAT-CANEGRO	1992–2007
Ramachandran et al. (2017)	Tamil Nadu, India	DSSAT-CANEGRO	1970–2000
Marin et al. (2015)	Piracicaba, Brazil	APSIM-Sugar, DSSAT-CANEGRO	1980–2010
Sonkar et al. (2020)	Uttar Pradesh, India	DSSAT-CANEGRO	1971–2000

Data collection and samples

From each of the nine eligible studies, we extracted two variables: the change in mean temperature relative to the baseline (ΔT , °C) and the corresponding percentage change in sugarcane yield (PCY, %). Each observation represented a single simulation output, i.e. a reported percent change in yield for a given study, site, crop model, imposed temperature increase (and other simulation variables where applicable). PCY was calculated as the difference between the simulated and baseline mean yields, divided by the baseline mean yield and expressed as a percentage. Where studies tested multiple temperature levels in sensitivity analyses without using a climate model, each temperature increment was treated as a separate observation, consistent with the approach used by Li et al. (2021).

From Table 1, DSSAT-CANEGRO was used in eight of the nine studies, while APSIM-Sugar and AgMIP GGCMI each appear in only one study, both at single sites. This limits the ability to separate crop-model effects from study-level effects. Furthermore, baseline periods span from 1960–1990 to 1981–2022, meaning the reference yield against which PCY is calculated reflects substantially different historical climates, as commonly observed in crop-climate meta-analyses (Li et al., 2021).

The sole predictor variable in this analysis is ΔT , as it was the only climate driver reported consistently across all nine studies. Other potentially relevant variables, such as changes in precipitation, atmospheric CO₂ concentration, climate models and scenarios, irrigation method and varieties, were available only for a subset of studies and could not be included without significantly reducing the number of observations. After excluding observations with missing values for either PCY or ΔT , the final dataset comprised 45 simulation observations.

Statistical analysis

From the collected data, we built a random effect regression model to predict the percent change in sugarcane yield (PCY, %) relative to each study's baseline period, using the temperature change applied in each simulation as predictor (ΔT , °C). A similar meta-analysis by Li et al. (2021) also included precipitation change, CO₂ concentration, and adaptation measures as fixed-effect moderators to predict yield change for cotton. However, they were omitted from

the model due to insufficient complete cases: less than half of the included studies reported precipitation change or CO₂ concentration, and none of them reported any adaptation measures.

Sampling variance was set to zero, which treats each simulation as a point estimate and gives them equal weight. This is because the primary studies are deterministic crop-model simulations, each reporting a single projected yield change without a standard error or replicated measurements, hence simulation-level sampling variance cannot be computed from the available data.

A multilevel meta-regression was fitted using `metafor::rma.mv` with REML estimation, with temperature change as the sole fixed moderator. Four random-effects structures were compared in a sequential manner:

1. **m_sim** (`~ 1 | sim`): a single-level model treating all 45 simulations as independent. This serves as a baseline reference that ignores the known clustering of multiple estimates within each study or their shared characteristics like study site or crop model.
2. **m_li** (`~ 1 | sim + ~ 1 | site + ~ 1 | crop_model`): a replication of the random-effects structure used by Li et al. (2021) for their cotton meta-analysis, which included study site and crop model as crossed random intercepts to model the dependence between simulations from the same location and crop model. Li's two additional random effects—climate model and emission scenario—could not be replicated here because approximately half of the sugarcane studies are sensitivity analyses that apply prescribed temperature increments without using a climate model or an emission scenario.
3. **m_full** (`~ 1 | study/sim + ~ 1 | site + ~ 1 | crop_model`): extends **m_li** by adding a hierarchical study-level random intercept. Multiple simulation outputs drawn from the same publication are not independent, yet this within-study dependence was not explicitly modelled by Li et al. (2021). The nested term `study/sim` partitions variance into a between-study component and a within-study component.
4. **m_final** (`~ 1 | study/sim + ~ 1 | crop_model`): obtained by dropping `site` from **m_full** after its variance component was estimated to effectively zero in the full model, indicating redundancy once study clustering was accounted for. The formula for the final model is below:

$$PCY_{ij} = \beta_1 \cdot \Delta T_{ij} + \beta_0 + a_{\text{crop_model}} + b_i + e_{ij}$$

- PCY_{ij} : percent yield change for simulation j in study i
- β_1 : fixed slope for temperature change
- ΔT_{ij} : temperature change (°C) for simulation j in study i
- β_0 : overall fixed intercept
- $a_{\text{crop_model}}$: random intercept for crop model; $N(0, \sigma_{\text{crop_model}}^2)$

- b_i : random intercept for study i ; $N(0, \sigma_{\text{between study}}^2)$
- e_{ij} : random intercept for simulation j in study i ; $N(0, \sigma_{\text{within study}}^2)$

To assess whether the temperature–yield relationship differed systematically across crop models or studies, two extensions were tested: a random slope across crop models, and a random slope across studies. A fixed $\Delta T \times$ crop model interaction was additionally considered as an alternative to the random-slope formulation, treating the crop model as a systematic differentiator of the temperature effect that warrants a separately reported slope for each model.

Results

Effect sizes

Out of the 45 simulated changes in sugarcane yield across 9 selected studies, 32 showed a decrease in yield (71%), while only 13 showed an increase (29%). The mean yield change across simulations is -6.6%, with a 95% confidence interval of [-11.1%, -2.1%], indicating significant yield loss on average. Lastly, yield tends to fall as temperature rises.

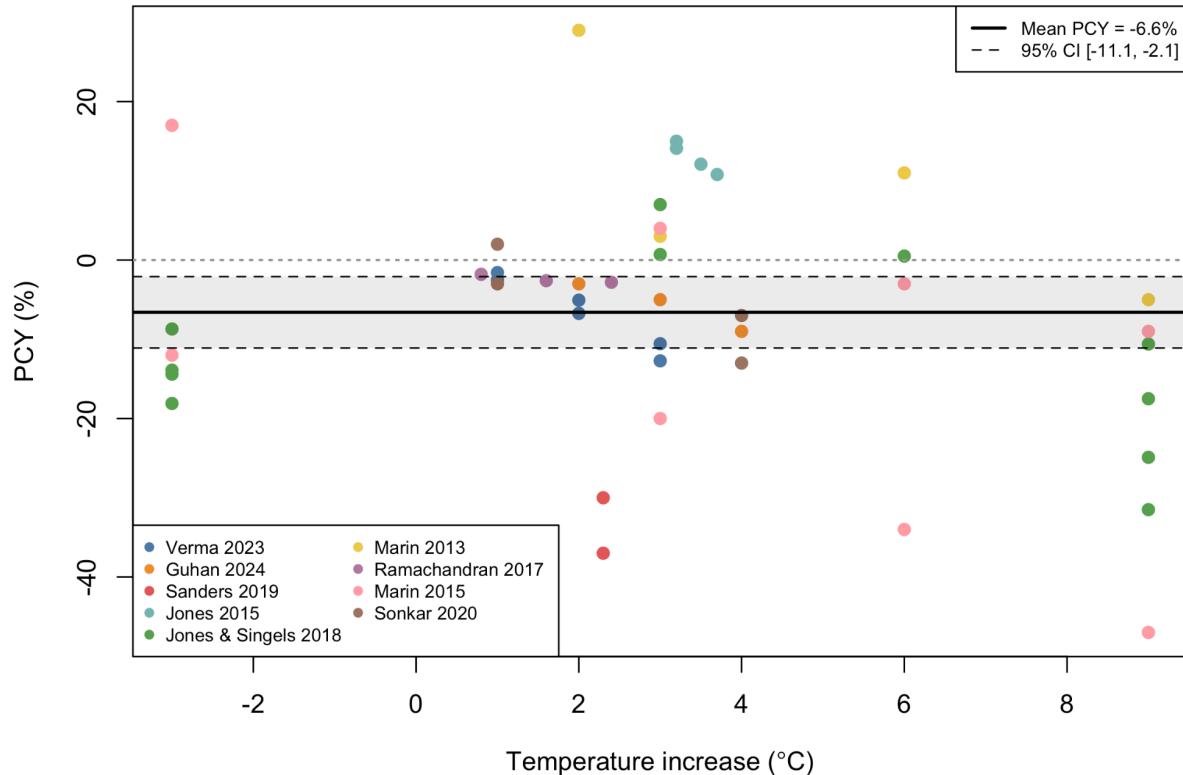


Figure 1: Percent change in sugarcane yield vs temperature increase across simulations

Model selection

Table 2 includes fit statistics for the four random-effects models to study the effect on sugarcane yield by temperature change—the fixed-effect moderator. Adding site and crop-model random intercepts to the baseline model significantly improved fit ($\Delta\text{AIC} = -4.4$). Introducing the hierarchical study-level intercept yielded a further significant improvement ($\Delta\text{AIC} = -5.5$), confirming that within-study dependence is a meaningful source of heterogeneity that the Li-style structure did not capture.

The site-level random effect was found to be redundant once the study-level intercept was included, as its variance component was estimated at zero. This redundancy likely stems from the fact that most individual studies are conducted at a single, unique location. Consequently, removing the site effect from the full model resulted in the most parsimonious fit, yielding the lowest AIC and BIC values across all candidate models.

Table 2. Estimates and fit for the random-effects models. **m_final** (bold) is the preferred model

Model	Random structure	AIC	Slope (%/°C)	95% CI	p
m_sim	simulation	359.2	-1.07	[-2.32, 0.17]	0.092
m_li	simulation + study site + crop model	354.8	-1.07	[-2.16, 0.01]	0.052
m_full	study/simulation + study site + crop model	349.3	-1.29	[-2.19, -0.38]	0.005
m_final	study/simulation + crop model	347.3	-1.29	[-2.19, -0.38]	0.005

Model estimates

Table 3 shows the fixed-effects and random-effects estimates of the final model. It includes a statistically-significant slope of -1.29 ($p = 0.005$), indicating that each additional degree of warming reduces simulated sugarcane yield by approximately 1.29 percentage points on average. In contrast, the model's intercept—estimated at -12.43 —does not reach statistical significance ($p = 0.188$). The large yet statistically insignificant intercept could be due to confounding with unmodelled drivers, such as CO_2 concentration and precipitation change, as well as an imprecise extrapolation at the intercept due to the natural absence of simulations with zero temperature change.

The standard deviations of the random-intercept effects are 8.09 PCY percentage points at the between-study level, 10.40 at the within-study level, and 14.22 across crop models, reflecting substantial heterogeneity at all levels.

Table 3: Final model fixed-effects and random-effects estimates

Fixed Effects

Parameter	Estimate	SE	z	p	95% CI
Intercept	-12.43	9.45	-1.32	0.188	[-30.95, 6.08]
ΔT (°C)	-1.29	0.46	-2.80	0.005	[-2.19, -0.38]

Random Effects (Variance Components)

Variance component	Variance (σ^2)	SD
Between-study	65.52	8.09
Within-study	108.07	10.40
Crop model	202.29	14.22

Figure 2 shows the fitted regression line overlaid on all 45 simulations, showing the negative relationship across the ΔT range covered by the studies (-3 °C to $+9$ °C).

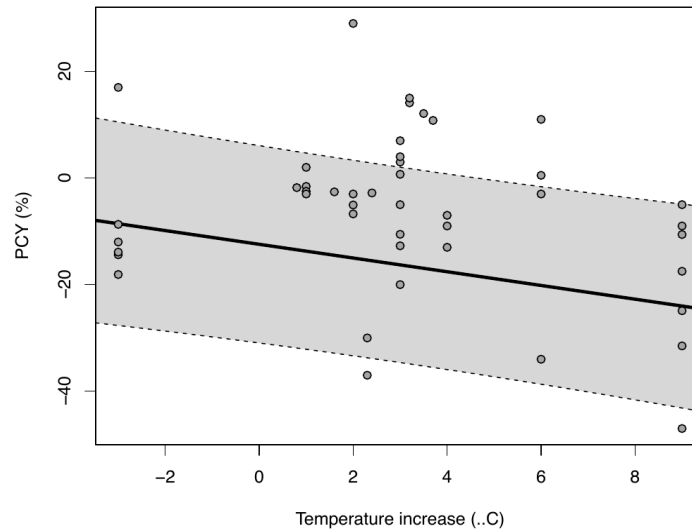


Figure 2: Fitted regression line overlaid on all 45 simulations

Model validation & sensitivity analysis

A normal quantile–quantile plot of the simulation-level standardized residuals from the final model (Figure 3) shows the points falling close to the diagonal line. This implies that the simulation-level residuals satisfy the normality assumption.

Influence measure for each simulation (standardized deleted residuals, Cook's distance, and leverage; Figure 4) identified simulation 34—an APSIM-Sugar run from Marin et al. (2015) with $\Delta T = -3^\circ\text{C}$ and a corresponding 17% yield increase—as a clear outlier: its Cook's distance of 1.6 is an order of magnitude larger than that of any other simulation, combining high leverage with a large residual.

As a result, the final model was fitted without this study for sensitivity analysis. The regression slope became markedly less negative and no longer significant: from $-1.29\%/^\circ\text{C}$ ($p = 0.005$) to $-0.78\%/^\circ\text{C}$ ($p = 0.082$) (Figure 5). While this could imply that the model's conclusion on the effect of temperature change on sugarcane yield is sensitive to outliers, this lack of statistical significance could also be contributed by the small number of observations after removing the outliers.

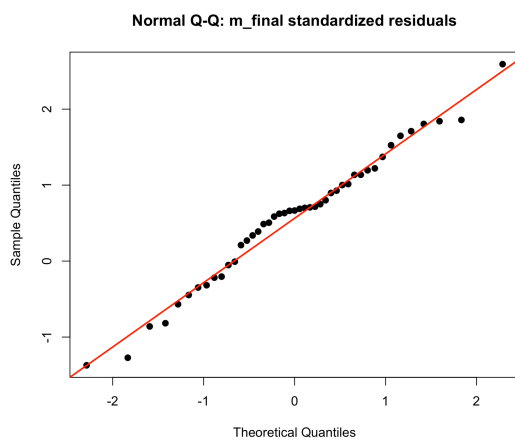


Figure 3: QQ plot of standardized residuals

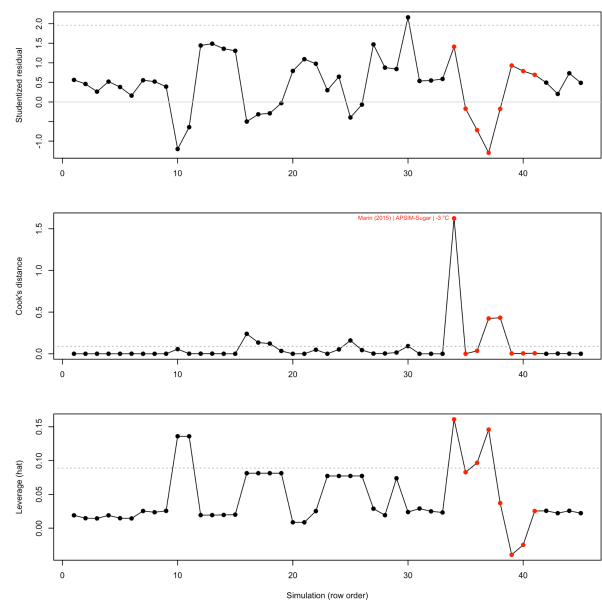


Figure 4: influence metrics for each simulation (Marin et al. (2015)'s highlighted in red)

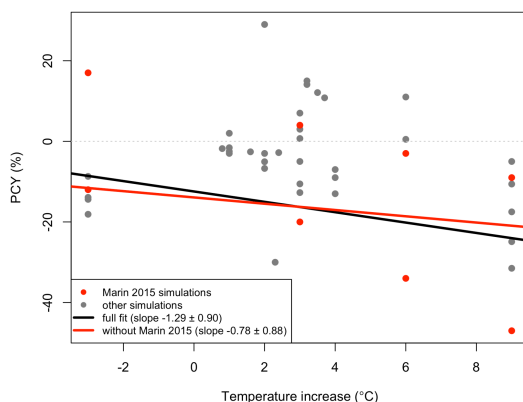


Figure 5: Regression line with and without Marin et al. (2015)

Model extensions

Figure 6 displays individual regression lines per study, and figure 7 displays regression lines per crop model, illustrating some between-study and between-model heterogeneity in slope. However, models with study-level or model-level random slopes were rejected: in each case, the estimated intercept–slope correlation reached the boundary of the parameter space ($\rho = \pm 1.00$), indicating a singular fit on the few observations from the selected studies.

A model with fixed effect of crop models on the slope was also not retained: AgMIP GGCM crop model has only two simulations at a single ΔT value, making its slope unable to estimate, and the APSIM-Sugar slope rested entirely on a single study (Marin 2015), confounding crop-model and study effects. The temperature slope was therefore kept fixed across all groups, and study and crop models only contribute to the random intercept of the model.

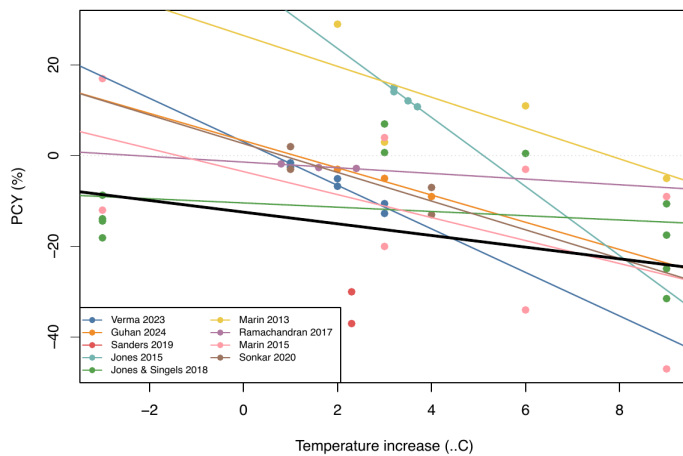


Figure 6: Regression line per study

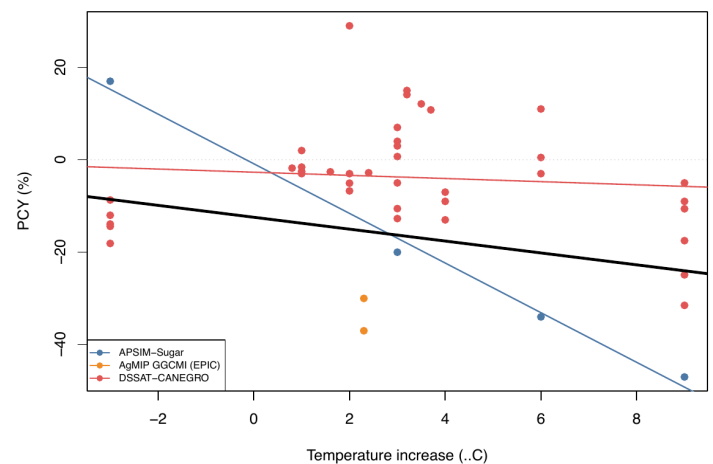


Figure 7: Regression line per crop model

Discussion

While the simulated yield change is negative on average (-6.6% , with 95% CI $[-11.1\%, -2.1\%]$), this mean conceals substantial heterogeneity: part of the spread is systematic—each additional degree of warming reduces simulated sugarcane yield by approximately 1.29 percentage points on average—but considerable residual variation remains across crop models (SD = 14.22), between studies (SD = 8.09), and between simulations within a study (SD = 10.40).

The meta-analytic estimate of temperature effect on sugarcane yield of $-1.29\%/^{\circ}\text{C}$ aligns with a broader trend of declining yields for crops as temperatures rise, such as a $-1.64\%/^{\circ}\text{C}$ response for cotton in Li et al. (2021). A methodological improvement over Li's meta-analysis is the explicit study-level random intercept, which sharpened the temperature slope from $-1.07\%/^{\circ}\text{C}$ ($p = 0.09$) to $-1.29\%/^{\circ}\text{C}$ ($p = 0.005$).

Several limitations affect the generalisability of the results. The nine included studies are geographically concentrated in India, Brazil, and southern Africa, with limited representation of other major sugarcane-producing regions such as Southeast Asia, China, and the Caribbean. Furthermore, the simulations are heavily dominated by the DSSAT-CANEGRO crop model. Therefore, the regression slope reflects the temperature sensitivity of sugarcane production primarily in those contexts and may not generalise globally and under different crop models.

Additionally, publication bias cannot be ruled out: simulation studies reporting noteworthy yield changes may be more likely to be published, potentially inflating the estimated temperature effect. The included studies in this meta-analysis reported single deterministic yield-change estimate from each simulation without uncertainty measures, so standard publication-bias diagnostics such as a funnel plot cannot be applied.

Conclusion

This meta-analysis synthesised 45 simulation outputs from nine studies to quantify how climate change affects sugarcane yield. Addressing our two research questions, we found that a majority of simulations (32 of 45, 71%) projected yield declines, with a mean change of -6.6% (95% CI -11.1 to -2.1), and that each additional 1°C of warming was associated with a reduction of roughly 1.3 percentage points in yield (-1.29%/°C, $p = 0.005$). Methodologically, adding an explicit study-level random intercept—absent from the Li et al. (2021) specification—accounted for within-study dependence and sharpened the temperature effect from -1.07%/°C ($p = 0.09$) to -1.29%/°C ($p = 0.005$).

At the same time, the result warrants caution: removing Marin (2015)—the source of several outliers—weakened the slope to -0.78%/°C and rendered it non-significant ($p = 0.082$). Furthermore, it reflects the temperature sensitivity of sugarcane production primarily in the dominating locations (India, Brazil, and southern Africa) and crop models (DSSAT-CANEGRO), and may not generalise globally and under different crop models.

Lastly, the analysis was also restricted to temperature, the only driver reported consistently across all nine studies, leaving out precipitation, CO₂ concentration, climate models & scenarios, and adaptation measures that Li et al. (2021) were able to include. Future work should therefore extend the systematic review to a broader and more geographically balanced set of studies and crop models—including gray literature (e.g., dissertations, technical reports, and unpublished simulation runs obtained by contacting active research groups). Extracting additional climate moderators from these studies will enable a richer multi-factor model and more nuanced conclusions about the drivers of sugarcane yield change under climate change.

Appendix

A.1 Published studies used in this analysis

1. Guhan, V., et al. (2024). Assessing the impact of climate change on water requirement and yield of sugarcane over different agro-climatic zones of Tamil Nadu. *Scientific Reports*, 14, 58771.
2. Jones, M.R., Singels, A., et al. (2015). Simulated impacts of climate change on water use and yield of irrigated sugarcane in South Africa. *Agricultural Systems*, 137, 20–30.
3. Jones, M.R. & Singels, A. (2018). Refining the Canegro model for improved simulation of climate change impacts on sugarcane. *European Journal of Agronomy*, 100, 76–86.
4. Marin, F.R., Jones, J.W., Singels, A., Royce, F., Assad, E.D., Pellegrino, G.Q., & Justino, F. (2013). Climate change impacts on sugarcane attainable yield in southern Brazil. *Climatic Change*, 117, 227–239.
5. Marin, F.R., Thorburn, P.J., Nassif, D.S.P., & Costa, L.G. (2015). Sugarcane model intercomparison: Structural differences and uncertainties under current and potential future climates. *Environmental Modelling & Software*, 72, 372–386.
6. Ramachandran, A., Praveen, D., Jaganathan, R., RajaLakshmi, D., & Palanivelu, K. (2017). Spatiotemporal analysis of projected impacts of climate change on the major C3 and C4 crop yield under representative concentration pathway 4.5: Insight from the coasts of Tamil Nadu, South India. *PloS one*, 12(7), e0180706.
7. Sanders, A., Thomas, T.S., Rios, A.R., & Dunston, S. (2019). *Climate Change, Agriculture, and Adaptation Options for Honduras*. IFPRI Discussion Paper 01883. Washington, D.C.: International Food Policy Research Institute.
8. Sonkar, G., Singh, N., Mall, R.K., Singh, K.K., & Gupta, A. (2020). Simulating the impacts of climate change on sugarcane in diverse agro-climatic zones of northern India using CANEGRO-Sugarcane model. *Sugar Tech*, 22(3), 460–472.
9. Verma, A. K., Garg, P. K., Prasad, K. H., & Dadhwal, V. K. (2023). Variety-specific sugarcane yield simulations and climate change impacts on sugarcane yield using DSSAT-CSM-CANEGRO model. *Agricultural Water Management*, 275, 108034.

A.2. Search strategy and PECO criteria

A comprehensive literature search was performed in the Web of Science and ScienceDirect databases using the following search string:

("sugarcane") AND ("climate change") AND ("drought" OR "frost" OR "heat" OR "flood" OR "water stress" OR "temperature" OR "extreme rainfall" OR "climate risk") AND (<country> OR <region>) AND ("yield")

With <country> and <region> are sugarcane-growing regions in South America, Central America and Mexico, South Asia, Southeast Asia and southern China, and Southern Africa.

The search was restricted to publications published in English between 2010 and 2025.

PECO eligibility criteria

Criterion	Definition
Population	Sugarcane crop in the target regions listed above
Exposure	Climate hazards or extremes (heat stress, frost, extreme/heavy rainfall, flood, drought, climate-driven waterlogging or water deficit stress) OR climate variables (temperature, precipitation, solar radiation)
Comparator	Baseline climate conditions, historical yields, or alternative climate scenarios
Outcome	Yield quantity, productivity, production, or biomass (stalk fresh or dry mass)

Studies were also required to employ statistical analyses, process-based crop model simulations, or a combination of both. Studies were excluded if their primary focus was genetic improvement, trait selection, or agronomic management without quantifying climate change impacts. Review papers, opinion pieces, and laboratory or greenhouse trials lacking field validation were also excluded.

A.3 PRISMA flow diagram

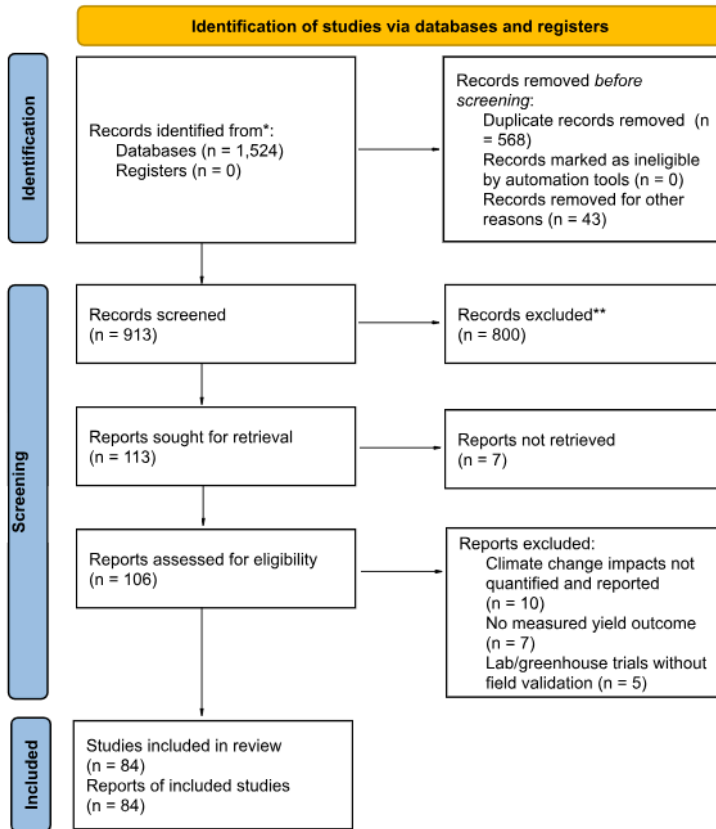


Figure A1. PRISMA flow diagram showing the screening process: records identified through database searching (Web of Science and ScienceDirect), records after duplicate removal, records screened by title and abstract, full-text articles assessed for eligibility, and studies included in the systematic review (n = 84). Of these, nine studies reporting temperature-yield data points from process-based crop simulation models were extracted for the meta-analysis.

A.4 Use of Generative AI

In preparing this report, we used the Cursor IDE assistant and Gemini as a coding and writing aid. Its use is also declared in the accompanying KU Leuven GenAI code-of-conduct form. It was used in three ways:

- **Code cleanup:** After we had written and run the full analysis in R, we used the assistant to refactor the script: removing exploratory and unused code, annotating each remaining block with the report table or figure it produces, and standardizing all plot outputs to PNG. Example prompt: "Clean up sugarcane_ma.R to match only the results shown in the report and remove code that doesn't add to it." We reviewed every change and re-ran the script to confirm the numerical results and figures were unchanged.
- **Concept explanations and cross-checking:** We asked the assistant to explain statistical concepts (e.g., omitted-variable confounding and the interpretation of a non-significant, extrapolated intercept) and to compare our draft against the course paper guidelines and the reporting lecture. Example prompt: "Compare my report with the grading guidelines and tell me what to improve." We verified its suggestions against the lecture slides and the cited literature and did not copy text from it for these parts.
- **Language assistance:** We used the AI to review, rephrase and improve the grammar, clarity and tone of passages we had drafted ourselves.

All analytical decisions, the model specification, the interpretation of the results, and the final text are our own, and we are able to explain them. We critically checked all AI output for correctness, consistent with the KU Leuven guidelines on responsible use of Generative AI.

A.5 Analysis code

```
library(metafor)

dat <- read.csv("ma_data_clean.csv", stringsAsFactors = FALSE)
dat <- dat[dat$include == "yes", ]
dat$PCY_pct <- as.numeric(dat$PCY_pct)
dat$delta_T_C <- as.numeric(dat$delta_T_C)
dat$study_id <- as.factor(dat$study_id)
dat <- dat[!is.na(dat$PCY_pct) & !is.na(dat$delta_T_C), ]
dat$sim <- factor(seq_len(nrow(dat)))
dat$vi <- 0

studies <- sort(unique(dat$study_id))
study_labels <- c(
  "65VYUJZ8" = "Verma 2023",
  "7PCFLLSK" = "Guhan 2024",
  "AADRMRIW" = "Sanders 2019",
  "FWLHHVEY" = "Jones 2015",
  "HP54EZKH" = "Jones & Singels 2018",
  "PCXRP7XW" = "Marin 2013",
  "Q6N2E27T" = "Ramachandran 2017",
  "XA3Z3WRK" = "Marin 2015",
  "Z5PF5XNQ" = "Sonkar 2020"
)
cols <- setNames(palette.colors(length(studies), "Tableau 10"), studies)

# Figure 1: Percent change in sugarcane yield vs temperature increase
across simulations
n_total <- nrow(dat)
n_below <- sum(dat$PCY_pct < 0)
n_above <- sum(dat$PCY_pct > 0)
n_zero <- sum(dat$PCY_pct == 0)
pct_below <- 100 * n_below / n_total
pct_above <- 100 * n_above / n_total
cat(sprintf("\nSimulations: n=%d | below 0: %d (0.1f%%) | above 0: %d
(0.1f%%) | at 0: %d\n",
          n_total, n_below, pct_below, n_above, pct_above, n_zero))
pcy_tt <- t.test(dat$PCY_pct)
pcy_mean <- unname(pcy_tt$estimate)
pcy_ci <- pcy_tt$conf.int
cat(sprintf("Mean PCY = 0.2f%% (95%% CI 0.2f to 0.2f)\n", pcy_mean,
pcy_ci[1], pcy_ci[2]))
```

```

png("ma_explore.png", width = 8, height = 6, units = "in", res = 300)
plot(dat$delta_T_C, dat$PCY_pct, pch = 19, col = cols[dat$study_id],
      xlab = expression("Temperature increase (" * degree * "C)"), ylab =
"PCY (%)")
usr <- par("usr")
rect(usr[1], pcy_ci[1], usr[2], pcy_ci[2], col = rgb(0, 0, 0, 0.08), border
= NA)
abline(h = pcy_mean, col = "black", lwd = 2)
abline(h = pcy_ci, col = "black", lwd = 1, lty = 2)
abline(h = 0, col = "grey60", lty = 3, lwd = 1.5)
legend("bottomleft", legend = study_labels[studies], col = cols[studies],
      pch = 19, cex = 0.7, ncol = 2, bg = "white")
legend("topright",
      legend = c(sprintf("Mean PCY = %.1f%%", pcy_mean),
                  sprintf("95% CI [%.1f, %.1f]", pcy_ci[1], pcy_ci[2])),
      lty = c(1, 2), lwd = c(2, 1), col = "black", cex = 0.7, bg =
"white")
dev.off()

```

Table 2: Estimates and fit for the random-effects models. m_final is the preferred model

```

m_sim <- rma.mv(yi = PCY_pct, V = vi, mods = ~ delta_T_C,
               random = list(~ 1 | sim),
               data = dat, method = "REML")
summary(m_sim)

m_li <- rma.mv(yi = PCY_pct, V = vi, mods = ~ delta_T_C,
               random = list(~ 1 | sim, ~ 1 | site, ~ 1 | crop_model),
               data = dat, method = "REML")
summary(m_li)
anova(m_sim, m_li)

m_full <- rma.mv(yi = PCY_pct, V = vi, mods = ~ delta_T_C,
                 random = list(~ 1 | study_id/sim, ~ 1 | site, ~ 1 |
crop_model),
                 data = dat, method = "REML")
summary(m_full)
anova(m_li, m_full)

m_final <- rma.mv(yi = PCY_pct, V = vi, mods = ~ delta_T_C,
                  random = list(~ 1 | study_id/sim, ~ 1 | crop_model),
                  data = dat, method = "REML")

```

```

anova(m_final, m_full)

data.frame(model = c("sim", "sim + site + crop", "study/sim + site +
crop", "study/sim + crop"),
          AIC = c(AIC(m_sim), AIC(m_li), AIC(m_full), AIC(m_final)),
          BIC = c(BIC(m_sim), BIC(m_li), BIC(m_full), BIC(m_final)),
          row.names = NULL)

# Table 3: Final model fixed-effects and random-effects estimates
summary(m_final)

# Figure 2: Fitted regression line overlaid on all 45 simulations
png("ma_regplot.png", width = 7, height = 6, units = "in", res = 300)
regplot(m_final, mod = "delta_T_C", psize = 1,
        xlab = "Temperature increase (\u00b0C)", ylab = "PCY (%)")
dev.off()

# Figure 3: QQ plot of standardized residuals
res_z <- rstandard(m_final)$z
png("ma_qqresid.png", width = 7, height = 6, units = "in", res = 300)
qqnorm(res_z, pch = 19, main = "Normal Q-Q: m_final standardized
residuals")
qqline(res_z, col = "red", lwd = 2)
dev.off()

# Figure 4: Influence metrics for each simulation (Marin et al. (2015)
highlighted in red)
res_stud <- rstudent(m_final)$z
cookd <- cooks.distance(m_final)
hatv <- hatvalues(m_final)
idx <- seq_along(res_stud)
cook_cut <- 4 / length(cookd)
flag <- which(abs(res_stud) > qnorm(0.975) | cookd > cook_cut)
out_i <- which.max(cookd)
out_lab <- sprintf("Marin (2015) | %s | %g \u00b0C",
dat$crop_model[out_i], dat$delta_T_C[out_i])
pt_col <- ifelse(dat$study_id == "XA3Z3WRK", "red", "black")
png("ma_influence.png", width = 8, height = 8, units = "in", res = 300)
op <- par(mfrow = c(3, 1), mar = c(4, 4.5, 1.5, 1))
plot(idx, res_stud, type = "o", pch = 19, col = pt_col, xlab = "", ylab =
"Studentized residual")
abline(h = c(-1, 1) * qnorm(0.975), col = "grey70", lty = 3)
abline(h = 0, col = "grey85")

```

```

plot(idx, cookd, type = "o", pch = 19, col = pt_col, xlab = "", ylab =
"Cook's distance")
abline(h = cook_cut, col = "grey70", lty = 3)
text(out_i, cookd[out_i], labels = out_lab, pos = 2, cex = 0.7, col =
"red")
plot(idx, hatv, type = "o", pch = 19, col = pt_col,
      xlab = "Simulation (row order)", ylab = "Leverage (hat)")
abline(h = 2 * mean(hatv), col = "grey70", lty = 3)
par(op)
dev.off()

infl_tab <- data.frame(row      = idx,
                      study_id = dat$study_id,
                      crop_model = dat$crop_model,
                      delta_T_C = dat$delta_T_C,
                      PCY_pct   = dat$PCY_pct,
                      stud_res  = round(res_stud, 2),
                      cooks_d   = round(cookd, 3),
                      hat       = round(hatv, 3))
cat("\nFlagged influential/outlying simulations (|stud.res|>1.96 or Cook's
D > ",
    round(cook_cut, 3), "):\n")
print(infl_tab[flag, ], row.names = FALSE)

# Figure 5: Regression line with and without Marin et al. (2015)
dat_nox <- droplevels(dat[dat$study_id != "XA3Z3WRK", ])
m_nox <- rma.mv(yi = PCY_pct, V = vi, mods = ~ delta_T_C,
               random = list(~ 1 | study_id/sim, ~ 1 | crop_model),
               data = dat_nox, method = "REML")
comp <- data.frame(
  term      = c("intercept", "delta_T_C"),
  full      = round(coef(m_final), 3),
  full_se   = round(m_final$se, 3),
  full_p    = round(m_final$pval, 4),
  nox       = round(coef(m_nox), 3),
  nox_se    = round(m_nox$se, 3),
  nox_p     = round(m_nox$pval, 4))
print(comp, row.names = FALSE)
cat(sprintf("n: full = %d sims / %d studies; nox = %d sims / %d studies\n",
            m_final$k, length(unique(dat$study_id)),
            m_nox$k, length(unique(dat_nox$study_id))))

pt_col <- ifelse(dat$study_id == "XA3Z3WRK", "red", "grey50")

```

```

mar_full <- (m_final$ci.ub[2] - m_final$ci.lb[2]) / 2
mar_nox <- (m_nox$ci.ub[2] - m_nox$ci.lb[2]) / 2
png("ma_regplot_nox.png", width = 7, height = 6, units = "in", res = 300)
plot(dat$delta_T_C, dat$PCY_pct, pch = 19, col = pt_col,
      xlab = expression("Temperature increase (" * degree * "C)"), ylab =
"PCY (%)")
abline(h = 0, col = "grey80", lty = 3)
abline(a = coef(m_final)[1], b = coef(m_final)[2], col = "black", lwd = 3)
abline(a = coef(m_nox)[1], b = coef(m_nox)[2], col = "red", lwd = 3)
legend("bottomleft", bg = "white", cex = 0.8,
      legend = c("Marin 2015 simulations", "other simulations",
sprintf("full fit (slope %.2f \u00b1 %.2f)",
coef(m_final)[2], mar_full),
sprintf("without Marin 2015 (slope %.2f \u00b1 %.2f)",
coef(m_nox)[2], mar_nox)),
      pch = c(19, 19, NA, NA), lwd = c(NA, NA, 3, 3),
      col = c("red", "grey50", "black", "red"))
dev.off()

# Model extensions
m_rslope <- tryCatch(
  rma.mv(yi = PCY_pct, V = vi, mods = ~ delta_T_C,
        random = list(~ 1 | study_id/sim, ~ delta_T_C | crop_model),
        struct = "GEN", data = dat, method = "REML"),
  error = function(e) e)
if (inherits(m_rslope, "error")) {
  cat("m_rslope problem:", conditionMessage(m_rslope), "\n")
} else {
  print(summary(m_rslope))
  print(anova(m_final, m_rslope))
}

m_sslope <- tryCatch(
  rma.mv(yi = PCY_pct, V = vi, mods = ~ delta_T_C,
        random = list(~ delta_T_C | study_id, ~ 1 | sim, ~ 1 |
crop_model),
        struct = "GEN", data = dat, method = "REML"),
  error = function(e) e)
if (inherits(m_sslope, "error")) {
  cat("m_sslope problem:", conditionMessage(m_sslope), "\n")
} else {
  print(summary(m_sslope))
  print(anova(m_final, m_sslope))
}

```

```

}

m_fixint <- tryCatch(
  rma.mv(yi = PCY_pct, V = vi, mods = ~ delta_T_C * crop_model,
    random = list(~ 1 | study_id/sim), data = dat, method = "REML",
  test = "t"),
  error = function(e) e)
if (inherits(m_fixint, "error"))
  cat("m_fixint problem:", conditionMessage(m_fixint), "\n") else
summary(m_fixint)

# Figure 6: Regression line per study
png("ma_studies.png", width = 8, height = 6, units = "in", res = 300)
plot(dat$delta_T_C, dat$PCY_pct, pch = 19, col = cols[dat$study_id],
  xlab = "Temperature increase (\u00b0C)", ylab = "PCY (%)")
abline(h = 0, col = "grey80", lty = 3)
for (s in studies) {
  ds <- dat[dat$study_id == s, ]
  if (length(unique(ds$delta_T_C)) >= 2) abline(lm(PCY_pct ~ delta_T_C,
ds), col = cols[s], lwd = 1.5)
}
abline(a = coef(m_final)[1], b = coef(m_final)[2], col = "black", lwd = 3)
legend("bottomleft", legend = study_labels[studies], col = cols[studies],
  pch = 19, lwd = 1.5, cex = 0.7, ncol = 2, bg = "white")
dev.off()

# Figure 7: Regression line per crop model
crop_models <- sort(unique(dat$crop_model))
cm_cols <- setNames(palette.colors(length(crop_models), "Tableau 10"),
crop_models)
png("ma_cropmodels.png", width = 8, height = 6, units = "in", res = 300)
plot(dat$delta_T_C, dat$PCY_pct, pch = 19, col = cm_cols[dat$crop_model],
  xlab = "Temperature increase (\u00b0C)", ylab = "PCY (%)")
abline(h = 0, col = "grey80", lty = 3)
for (cm in crop_models) {
  dc <- dat[dat$crop_model == cm, ]
  if (length(unique(dc$delta_T_C)) >= 2) abline(lm(PCY_pct ~ delta_T_C,
dc), col = cm_cols[cm], lwd = 1.5)
}
abline(a = coef(m_final)[1], b = coef(m_final)[2], col = "black", lwd = 3)
legend("bottomleft", legend = crop_models, col = cm_cols,
  pch = 19, lwd = 1.5, cex = 0.7, bg = "white")
dev.off()

```

References

- Asseng, S., Ewert, F., Rosenzweig, C., Jones, J. W., Hatfield, J. L., Ruane, A. C., ... & Wolf, J. (2013). Uncertainty in simulating wheat yields under climate change. *Nature climate change*, 3(9), 827-832.
- Borenstein, M., Hedges, L. V., Higgins, J. P., & Rothstein, H. R. (2021). *Introduction to meta-analysis*. John Wiley & Sons.
- Challinor, A.J., Watson, J., Lobell, D.B., Howden, S.M., Smith, D.R., & Chhetri, N. (2014). A meta-analysis of crop yield under climate change and adaptation. *Nature Climate Change*, 4, 287–291. DOI: 10.1038/nclimate2153.
- FAO (2023). *World Food and Agriculture – Statistical Yearbook 2023*. Rome: Food and Agriculture Organization of the United Nations.
- Li, N., Yao, N., Li, Y., Chen, J., Liu, D., Biswas, A., ... & Chen, X. (2021). A meta-analysis of the possible impact of climate change on global cotton yield based on crop simulation approaches. *Agricultural Systems*, 193, 103221.
- Marin, F. R., Jones, J. W., Singels, A., Royce, F., Assad, E. D., Pellegrino, G. Q., & Justino, F. (2013). Climate change impacts on sugarcane attainable yield in southern Brazil. *Climatic change*, 117(1), 227-239.
- Marin, F. R., Thorburn, P. J., Nassif, D. S., & Costa, L. G. (2015). Sugarcane model intercomparison: Structural differences and uncertainties under current and potential future climates. *Environmental Modelling & Software*, 72, 372-386.
- Page, M.J., et al. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ*, 372, n71.
- White, J.W., Hoogenboom, G., Kimball, B.A., & Wall, G.W. (2011). Methodologies for simulating impacts of climate change on crop production. *Field Crops Research*, 124(3), 357–368.
- Zhao, D. & Li, Y.-R. (2015). Climate change and sugarcane production: Potential impact and mitigation strategies. *International Journal of Agronomy*, 2015, 547386.

Code of conduct and transparency statement on the use of GenAI for KU Leuven-students (academic year 2025-2026)

Generative AI (GenAI) assistance tools can be used to generate various types of content, including text, images, code, video, music, or combinations thereof. Common examples of such tools include ChatGPT, Google Gemini, Microsoft Copilot, Midjourney, Claude.ai, Perplexity.ai, and DALL-E, among others.

This code of conduct is a tool that helps students to be transparent about the use of GenAI and fits within the university's principles on academic integrity.

Important guidelines and remarks

- **Sensitive or personal data:** Some GenAI tools protect your input and sensitive or personal data better than others. There is often no transparency on what the owners of the AI applications do with the data entered. Therefore, **do not enter sensitive or personal data in free GenAI tools**. More info about what sensitive and personal data are, is described in the [three confidentiality levels of the KU Leuven data classification model](#) (non-confidential, confidential, strictly confidential). For confidential data you should preferably use **M365 Copilot Chat with your KU Leuven account**. If you use another GenAI tool, you must be **absolutely certain** it does **not store or reuse the data you enter**. **Try to avoid entering these data. Do so only if strictly necessary, and only to the extent required. For personal data, work with anonymous or pseudonymized data.** Additionally, in case of strictly confidential data this should first be discussed with the teaching staff of the course or your thesis supervisor.
- **Copyrighted materials:** For **lawfully obtained copyrighted material** you should preferably use **M365 Copilot Chat with your KU Leuven account**. If you use another GenAI tool, you must be **absolutely certain** it does **not store or reuse the data you enter**.
- GenAI assistance may not be used for data or topics covered by a [Non-Disclosure Agreement \(NDA\)](#). Even in the absence of an NDA, certain information may still need to be treated as confidential—for example, due to regulatory requirements or the risk of significant harm to the university if disclosed. In case of doubt, check with your teaching staff or supervisor.
- If your master's thesis is under [embargo](#), you should first discuss with your supervisor whether the use of GenAI is permitted.
- Before using a GenAI tool, always consider whether its use is responsible, including from a [sustainability perspective](#) (e.g. when using GenAI as a search engine, language assistant,...).
- **Take a scientific and critical attitude** when interacting with GenAI assistance and interpreting its output, that may not always be correct.
- As a student you are responsible for complying with Article 84 of the Regulations on Education and Examinations: your report or thesis should reflect your own knowledge, understanding and skills. Be aware that plagiarism rules also apply to (work that is the result of) the use of GenAI assistance tools.

Exam Regulations Article 84: *“Every conduct individual students display with which they (partially) inhibit or attempt to inhibit a correct judgement of their own knowledge, understanding and/or skills or those of other students, is considered an irregularity which may result in a suitable penalty. A special type of irregularity is plagiarism, i.e. copying the work (ideas, texts, structures, designs, images, plans, codes , ...) of others or prior personal work in an exact or slightly modified way without adequately acknowledging the sources. Every possession of prohibited resources during an examination (see article 65) is considered an irregularity.”*

- In order to maintain academic integrity and avoid plagiarism, **more information about being transparent on the use of GenAI assistance and about correctly citing and referencing GenAI** can be found on this website for students_([Dutch](#)/[English](#)).
- **Additional reading: KU Leuven guidelines on responsible use of Generative AI tools, and other information** ([Dutch](#)/[English](#))

A few final words

If you are uncertain whether or not you should declare your use of GenAI tools, we suggest that you discuss this with your instructor or supervisor. It is always safer to declare GenAI use, even when it is not strictly required. However, declaring GenAI use does not entail that its use is allowed; the right column in the table below provides more detailed instructions in this regard (code of conduct).

Moreover, advanced AI tools are evolving rapidly, and their capabilities have expanded significantly in a short period of time. As a result, we do not yet have all the answers about their responsible use. Finally, it is important to follow-up on the most recent evolutions in AI technologies, to have an open mind but also to be a bit cautious, to communicate with instructors, teaching assistants, supervisors and peers, to be as transparent as we can, and to learn together as we move along.

Student name: Duy Khánh Nguyễn

Student number: r0968831

Please indicate with "X" whether it relates to a course assignment or to the Bachelor's or Master's thesis:

X This form is related to a **course assignment**.

Course name: Meta Analysis

Course number: G0B75A

This form is related to **my** O Bachelor's or O Master's thesis.

Title Bachelor's or Master's thesis:

Supervisor:

Daily supervisor:

Please indicate with "X":

O I **did not use** any GenAI assistance tool.

x I **did use** GenAI Assistance. In this case **specify which ones** (e.g. ChatGPT, M365 Copilot,...): Cursor, Gemini in Google Docs

GenAI assistance used as/for:	Name of the GenAI tool(s) used. If helpful, also describe in which way you were using GenAI related to what is specified as code of conduct.	Code of conduct:
		For each of the categories below, always take into account the important guidelines and remarks mentioned above (e.g. copyrighted data, sensitive or personal data,...).
As a language assistant for reviewing or improving texts I wrote myself	x	This use is similar to using spelling and grammar check tools. In general, you do not have to refer to such kind of GenAI use in the text. However, be careful: - When using GenAI tools on texts you did not write yourself to improve the text, you have to refer to the original source or author, otherwise you are committing plagiarism and thus an irregularity.
As a paraphrasing tool	x	This use is allowed except when it is prohibited by the teacher or the program of study. You may paraphrase your own text or texts by an author other than yourself and take inspiration from what a GenAI tool or another tool suggests (unless it is not allowed). In general, you do not have to refer to such kind of GenAI use or other paraphrasing tools in the text. However, be careful: - If it entails text by an author other than yourself, you are not allowed to include that paraphrased text without reference to the original source or author. Without such reference, you would be committing plagiarism and thus an irregularity.

For translation aid to improve texts I wrote myself or to better understand text from others		This use is allowed except when it is prohibited by the teacher or the program of study. It is similar to using translation tools (Google translate, DeepL, ...). In general, you do not have to refer to such kind of GenAI use in the text. However, be careful: - You are not allowed to include that translated text without reference to the original source or author. Without such reference, you would be committing plagiarism and thus an irregularity. - Always check the translated text for correctness and meaning.
As a search engine to get information on a topic or to search for existing research on the topic	x	This use is similar to e.g. a Google search or checking Wikipedia. If you write your own text based on this information, you do not have to refer to the use of GenAI in the text. You only have to refer to the existing research and references you have checked and used (without such references you would be committing plagiarism and thus committing an irregularity). However, be careful: - Be aware that the output of the GenAI tool cannot be guaranteed as a 100% reliable source of information. The output may not be entirely correct and/or be limited due to the databases it uses. Moreover, knowledge evolves and may change over time; therefore, the database of the GenAI tool may not be up to date. Therefore, verify the information and do not just copy-paste it as you should understand and critically process everything you are writing.
For literature search		This use is comparable to e.g. a Google Scholar search. You do not have to refer to such kind of GenAI use; you only have to refer to the literature references you have checked and used (without

		such references you would be committing plagiarism and thus committing an irregularity).. However, be careful: - Be aware that the search output is restricted to the database the GenAI tool is built on. After this initial search, look for scientific sources and conduct your own analysis of the source documents. Interpret, analyse and process the information you obtained; verify it and do not just copy-paste it as you should understand and critically process everything you are writing. - Be aware that some GenAI tools may output no or wrong references. As a student you are responsible for further checking and verifying the absence or correctness of references; do not just copy-paste it.
For generating programming code	x	Use of GenAI for coding is allowed except when it is prohibited by the teacher or the program of study. If used for coding, correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparant about the use of GenAI.
For generating new (research) ideas		This use of GenAI is allowed except when it is prohibited by the teacher or the program of study. Correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparant about the use of GenAI. Be careful: - Further verify in this case whether the idea is novel or not. It is likely that it is related to existing work. If so, that existing work should be correctly referenced in the text (without such reference you would be committing plagiarism and thus committing an irregularity). -

For generating synthetic data		Use of GenAI for generating synthetic data is allowed, provided that it is methodologically and ethically justifiable, except when it is prohibited by the teacher or the program of study. Always correctly mention the use of GenAI assistance in accordance with the instructions on the page on being transparant about the use of GenAI. Be careful: - Always carefully evaluate the generated synthetic data for quality and possible bias since the output is highly dependent on the quality of the data on which the models are trained.
For generating blocks of text (other than the allowed use without referencing mentioned above)		According to Article 84 of the Regulations on Education and Examinations your text should allow to correctly and properly assess your own knowledge, understanding and skills. Therefore, inserting blocks of text without quotes and a reference to GenAI assistance in your work is not allowed. Be careful: - If it is really needed to insert a block of text from a GenAI tool, for instance because of the nature of your assignment, mention it as a citation by using quotes and correctly mention the use of GenAI assistance in accordance with the instructions page on being transparant about the use of GenAI. - However, in general, such GenAI use should be kept to an absolute minimum; you should always check the original sources.
For generating visuals, video or audio		Use of GenAI for generating visuals, audio or video is allowed except when it is prohibited by the teacher or the program of study. Be careful:

		- If used, refer to GenAI for visuals according to the usual referencing style, following the instructions on the page on referencing GenAI. - If you work with existing visuals, audio or video, that existing work should be correctly referenced in the text (without such reference you would be committing plagiarism and thus an irregularity). - Explain the usage in the methods section (if there is one) and optionally attach (or link to) the prompts with the full output (or history).
Other use (specify here; this may also include a combination of types of use mentioned above):		To make sure other use of GenAI is allowed within the course or thesis, and if so, the conditions that may apply, contact the teaching staff of the course or the supervisor of the thesis beforehand and explain the intended GenAI use. Also inform the programme director. Motivate how you would comply with Article 84 of the exam regulations. Explain the use and the added value of the AI tool you consider to use and how it is in accordance with the assignment or thesis and the KU Leuven guidelines on responsible use of Generative AI tools. Depending on the kind of GenAI use, it may be needed to properly reference it in the text, in accordance with the instructions page on being transparant about the use of GenAI.